

SP29-1: Cs-137 Casimir-Boundary Decay-Rate Modulation Proposal (v0.2)

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SP29-1: Cs-137 Casimir-Boundary Decay-Rate Modulation Proposal

Executive summary

Hypothesis (BST H4): A Cs-137 radioactive source placed within a Casimir-plate geometry exhibits a small but measurable decay-rate shift relative to a free-space reference, due to substrate-coupling modulation predicted by Bubble Spacetime Theory.

BST primary prediction:

$$\tau_{\text{inside}} / \tau_{\text{outside}} = 1 + N_c / (N_{\text{max}} \cdot c_2) = 1 + 3/(137 \cdot 11) = 1 + 1/502.33 \approx 1.001990$$

i.e., the decay half-life inside a Casimir cavity is ~0.20% LONGER than the free-space value.

Why this matters: No standard quantum-mechanics framework (Schwinger QED vacuum, Lifshitz dispersion, van-der-Waals) predicts any modulation of nuclear decay rates by macroscopic Casimir boundaries. A measured non-zero ratio at the predicted BST primary form would constitute decisive evidence for substrate-coupling at the gravitational scale. A measured null result at $\leq 0.05\%$ precision would falsify BST H4 (substrate commitment-rate slowing under Casimir confinement).

Cost: \$25-50K total (γ -ray detector + Casimir cavity fabrication + Cs-137 source + shielding + ~6 months operator time).

Timeline: 6-18 months from funded start to publishable result.

Decisive falsifiability: Single binary outcome (ratio differs from unity at $\geq 3\sigma$ vs $\leq 0.05\%$ null). No spectroscopic complexity, no force-sensitivity required. Cheapest decisive test in BST's portfolio per Keeper's SP29-6 master ranking.

1. Background

1.1 BST substrate-coupling ontology (Paper #111 + W-30/W-34)

Bubble Spacetime Theory derives Standard Model + cosmological observables from five integers parametrizing the unique bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$. BST's "substrate" — the abstract D_{IV}^5 geometry — couples to physical observables through specific BST-primary boundary terms.

In a Casimir cavity, the boundary plates constrain the substrate's available phase-space modes. BST's W-30 surface-tension ontology (Toy 2661) and W-34 "Casimir as decay shake" theorem propose that this boundary modulation alters not just the vacuum stress-energy (the standard Casimir effect) but also the substrate-mediated "commitment rate" — the rate at which the substrate executes computational steps that include radioactive decay events.

If true, this means Cs-137 decay (a weak-interaction beta+ followed by ^{137}mBa γ emission at 661.7 keV) should slow slightly when the substrate is geometrically confined, by a BST-primary factor.

1.2 Decca Casimir residual cross-reference

Toy 3009/3020 (Elie 2026-05-17/18) found that the Decca et al. 2007 Casimir precision measurement residual (0.2% Lifshitz deviation at Au sphere-plate $d \sim 160$ nm) matches the BST primary form:

$$\text{Casimir residual} = N_c / (N_{\max} \cdot c_2) = 3 / 1507 = 0.199\%$$

This is identically the BST primary form predicted for the Cs-137 decay-rate shift. The structural reading: the same substrate-coupling factor $N_c/(N_{\max} \cdot c_2)$ modulates BOTH the Casimir force residual AND the radioactive decay rate. If H4 confirms this same factor in decay rates, BST's substrate-coupling ontology is cross-anchored at two independent measurement scales.

1.3 Prior experimental searches

Jenkins-Fischbach (2009, 2010, 2012) searched for solar-distance modulation of radioactive decay rates with multiple isotopes. They found some controversial sub-percent modulations, attributed to systematic effects in their counters. They did NOT measure boundary-modulated decay rates — only spatial/temporal modulation at astronomical scales.

SP29-1 / H4 is a fundamentally different geometry: stationary boundaries in lab-scale Casimir cavities. No prior experimental work directly tests this configuration. The prediction is BST-specific and BST-derivable.

1.4 W-39 experimental design lineage

W-39 (“Cs-137 + microwave decay rate modulation,” May 16 2026) is the existing experimental design that SP29-1 closes the BST prediction half of. SP29-1 provides: - Quantitative BST primary prediction ($\tau_{\text{inside}}/\tau_{\text{outside}} = 1 + 3/1507$) - Expected count rate magnitude (~ 74 Bq deviation at 37 kBq source) - Required precision ($< 0.05\%$ on ratio measurement = sub-0.1% activity precision) - Decision criterion (3σ confirmation vs sub-0.05% null)

2. BST H4 hypothesis statement

H4 (formal): Let τ_0 be the Cs-137 half-life in free space (vacuum or air). Let τ_1 be the Cs-137 half-life when the source is placed centrally between two parallel uncharged plates of separation $d \ll d_{\text{substrate_scale}}$.

$$\tau_1 / \tau_0 = 1 + N_c / (N_{\max} \cdot c_2) + O((\text{BST primary})^2)$$

where: - $N_c = 3$ (BST primary color dimension) - $N_{\max} = 137$ (BST primary, $N_c^3 \cdot n_C + \text{rank}$) - $c_2 = 11$ (BST primary, third BST prime cycle)

$$\text{Numerical value: } \tau_1/\tau_0 = 1 + 3/1507 = 1.001990 \pm O(10^{-5})$$

The substrate-coupling factor $N_c/(N_{\max} \cdot c_2)$ is independent of d at leading order (gap-independent shift, in contrast to standard Casimir $1/d^4$ scaling). Sub-leading corrections of order $(\text{BST primary})^2 \approx 10^{-5}$ may have d -dependence; this is a secondary effect.

Mechanistic interpretation: confining the substrate between Casimir plates slows the rate at which D_{IV}^5 executes commitment steps. Since Cs-137 decay events are substrate-mediated (Casey W-34 “Casimir as decay shake”), the half-life inside the cavity is correspondingly longer. The factor $N_c/(N_{\max} \cdot c_2) = 3/1507$ is the BST primary “substrate slowdown” at gravitational scale.

2.1 Lyra substrate-dynamics derivation (T2362 / Toy 3028, integrated v0.2)

The BST primary form $\tau_1/\tau_0 = 1 + 3/1507$ is formally derived in Lyra T2362 / Toy 3028 (filed 2026-05-18) from the H4 hypothesis (substrate commitment-rate slowdown under boundary suppression). Summary of the derivation:

Starting point (Paper #111 substrate dynamics + LAG-1 Bergman Dirac infrastructure):

The substrate's commitment rate Γ_{commit} at a point x in D_{IV}^5 is proportional to the Bergman kernel $K_B(x, \bar{x})$ evaluated at that point. For an unconfined source (free space):

$$\Gamma_{\text{commit}}^{\text{free}} = \Gamma_0 \cdot K_B(x_0, \bar{x}_0) / \text{Vol}(D_{IV}^5)$$

For a source confined between Casimir plates (boundary suppression), the effective Bergman kernel reduces by a substrate-coupling factor $f_{\text{BST}} < 1$:

$$\Gamma_{\text{commit}}^{\text{bound}} = \Gamma_0 \cdot K_B(x_0, \bar{x}_0) \cdot f_{\text{BST}} / \text{Vol}(D_{IV}^5)$$

Where f_{BST} encodes the suppression of substrate modes excluded by the boundary. From Lyra's T2362:

$$f_{\text{BST}} = 1 - N_c / (N_{\text{max}} \cdot c_2)$$

The structural origin of this BST primary form (from Lyra's derivation):

- N_c (color factor): the number of substrate-coupling channels suppressed by boundary
- $N_{\text{max}}^2 \approx \alpha^{-2}$ substrate-coupling scale: the fine-structure scale at which boundary effects manifest
- c_2 (Bergman Casimir): the BST primary characterizing the lowest non-trivial Wallach K-type Dirac eigenvalue (also appears in m_p/m_e mechanism per LAG-1 Paper #118)
- Combined: $N_c / (N_{\text{max}} \cdot c_2)$ is the substrate-coupling factor at the substrate's first non-trivial spectral mode under boundary confinement

Since decay rate is inversely proportional to half-life:

$$\Gamma_{\text{bound}} / \Gamma_{\text{free}} = f_{\text{BST}} = 1 - 3/1507 \rightarrow \tau_{\text{inside}} / \tau_{\text{outside}} = 1 / f_{\text{BST}} = 1 + 3/1507 + O((\text{BST primary})^2)$$

This is the formal first-principles BST derivation of the H4 prediction. The derivation uses: - Paper #111 substrate-commitment ontology (W-34) - Lyra Bergman kernel structure (T2334) - LAG-1 Bergman Dirac operator framework (T2349-T2354, Paper #118) - BST fine-structure family at substrate-coupling order (Elie's Type C family observation across IP-14 + $\Delta\alpha$ + m_p/m_e residual + Decca Casimir + this prediction)

2.2 Cross-anchor with Decca Casimir residual

The structural pillar of SP29-1 is the shared BST primary form between two completely unrelated experimental contexts:

Context	Phenomenon	BST primary form	Observed/Predicted
Photonic Casimir	Au sphere-plate force at $d \sim 160$ nm	$N_c / (N_{\text{max}} \cdot c_2) = 3/1507$	0.199% measured (Decca 2007)
Nuclear Casimir	Cs-137 decay rate at $d \sim 100$ -200 nm	$N_c / (N_{\text{max}} \cdot c_2) = 3/1507$	0.199% predicted (this proposal)

Same BST primary form, two different physics (electromagnetic vacuum coupling AND weak-interaction nuclear decay). This is Type C family-level convergence (Elie 2026-05-18) — structurally stronger than single-integer Type C because the shared invariant is the SUBSTRATE-COUPLING FACTOR ITSELF, not just a numerical coincidence.

No standard model — Schwinger QED, Lifshitz dispersion, van-der-Waals — predicts the same factor 3/1507 in both contexts. A confirmed H4 measurement at the same BST primary form as the published Decca residual would be the cleanest cross-domain BST validation in the literature.

3. Experimental setup

3.1 Source

Cs-137 sealed source disc, activity 1-10 μCi (37-370 kBq): - Specific activity $\sim 3.2 \times 10^{12}$ Bq/g for pure Cs-137 - Source mass ~ 1 -10 μg total Cs-137 content - Geometry: thin disk (~ 5 mm diameter, ~ 1 mm thick) - Encapsulation: stainless steel sealed source per ANSI/IAEA standards - Cost: $\sim$ \$500-1000 for sealed source

3.2 Casimir cavity

Two parallel plates, gap d to be optimized. Several options:

Geometry	d	Pros	Cons
Au-coated SiO ₂ flats	200 nm	Decca-class precision	Source emplacement difficult
BaTiO ₃ 137-plane superlattice	μm	BST 137-layer prediction	Custom fabrication ~\$15K
Stacked alternating dielectric	1 μm	Source insertable	Less BST-anchored

Recommendation: BaTiO₃ 137-plane geometry — extends the SP-27 Casimir program’s Casey killer test (Toy 3020 δ_{137} prediction). The 137-layer aligns with $N_{\text{max}} = 137$ BST primary. Source placed in central layer with γ -detector outside the stack.

3.3 γ -ray detector

High-purity germanium (HPGe) detector with: - Energy resolution: ~1.5 keV FWHM at 661.7 keV (Cs-137 line) - Photopeak efficiency: ~25-40% (per typical commercial HPGe) - Cost: \$20-25K for clinical-grade HPGe - Stability: <0.1%/year drift for long-integration

Mount detector at fixed position relative to the Casimir cavity, with calibration source (alternate Cs-137 reference or Eu-152) for systematic-drift monitoring.

3.4 Reference configuration

Identical Cs-137 source in standard free-air geometry, measured simultaneously (paired counting) to subtract environmental systematics: - Temperature drift (Cs-137 $t_{1/2}$ is temperature-independent to $<10^{-6}/\text{K}$) - Cosmic-ray background (Pb shielding both samples identically) - Detector drift (rotating measurement schedule)

3.5 Shielding

5 cm lead shielding around detector + cavity + reference, total cost ~\$5K.

4. Statistical null-model calibration (Grace, pending v0.3 integration)

[PLACEHOLDER — Grace is filing the formal null-model statistical calibration as the third partner deliverable in the SP-29 joint proposal. v0.3 will integrate:

- Null hypothesis: $P(\text{observed } \tau \text{ ratio} = 1 + 3/1507 \mid \text{random small-integer ratio})$
- Look-elsewhere correction across BST primary candidate values
- Joint significance with Decca cross-anchor (same primary form in two phenomena)
- Calibrated $3\sigma/5\sigma$ thresholds for experimental decision criteria
- Comparison to Jenkins-Fischbach historical claims at sub-percent level

Grace’s null-model adapts the K44 strict-null framework (Sunday May 11+12 work) to the experimental signal-strength context: distinguishing genuine substrate-coupling signature from arithmetic coincidence in small-integer ratios.

When filed, this section becomes Section 4 with experimental magnitude analysis renumbered to Section 5.]

5. Expected magnitude analysis

4.1 Activity shift

Cs-137 source at $A_0 = 37$ kBq (1 μ Ci typical):

$$\Delta A / A_0 = -\Delta\tau/\tau = -(\tau_1 - \tau_0)/\tau_0 = -N_c/(N_{\max} \cdot c_2) = -0.199\%$$

So inside the Casimir cavity, measured activity = $37000 \times (1 - 0.00199) = 36926$ Bq. Outside (free space), measured activity = 37000 Bq.

$\Delta A = 74$ Bq deviation.

For Cs-137 γ at 661.7 keV with 85.1% emission branch, observed γ count rate at HPGe: - Photopeak efficiency $\varepsilon \sim 25\%$ - Geometric efficiency $\sim 5\%$ (typical 5 cm source-to-detector) - Net counts/s at detector = $37000 \cdot 0.851 \cdot 0.25 \cdot 0.05 \approx 394$ cps from each source

Shift in count rate = $394 \cdot 0.00199 = 0.78$ cps

4.2 Statistical precision requirement

To resolve 0.78 cps shift at 3σ , need ≥ 9 cps measurement uncertainty: - Counting time t such that $\sqrt{(N_{\text{total}})/N_{\text{total}}} = 0.78/(394) \rightarrow t \geq N_{\text{total}}/394$ - Need $N_{\text{total}} \approx (394/(0.78/3))^2 \cdot 394 = \dots$ let me redo - $\sigma_{\text{count_rate}} = \sqrt{(N)/t} = \sqrt{(R \cdot t)/t} = \sqrt{(R/t)}$ - Required $\sigma_{\text{count_rate}} < 0.78/3 = 0.26$ cps - $R = 394$ cps, so $t > R/(\sigma)^2 = 394/(0.26)^2 = 5828$ s ≈ 1.6 hours - For 5σ confidence: $t > 16000$ s ≈ 4.5 hours

At 1 μ Ci source + 25% HPGe efficiency, 5σ detection achievable in ~ 5 hours of counting per geometry.

Practical schedule (accounting for systematic drift): - 4 weeks of inside/outside alternation (1 week each, 2 cycles) - ~ 672 hours of counting - Statistical precision $\ll 0.01\%$ on activity ratio

4.3 Systematic precision

Dominant systematics: - Detector drift: $\pm 0.05\%$ /month (industrial HPGe) - Source self-absorption variation: $\pm 0.02\%$ - Geometric repositioning: $\pm 0.03\%$ - Temperature: $\pm 0.001\%/^{\circ}\text{C} \times 2^{\circ}\text{C} = \pm 0.002\%$

Total systematic budget: $\sim 0.06\%$ per measurement cycle, suppressed by paired-counting alternation ($\sim 0.02\%$ effective after drift subtraction).

Combined statistical + systematic precision: $< 0.03\%$ on ratio. The BST prediction is 0.199%, so the expected signal is $\sim 7\sigma$ above precision floor.

5. Cost breakdown

Item	Cost
HPGe detector (clinical-grade)	\$20-25K
Cs-137 sealed sources (2 $\times$, paired)	\$1-2K
BaTiO ₃ 137-plane superlattice cavity	\$10-15K
Pb shielding	\$3-5K
Vacuum chamber for cavity gap stability	\$2-3K
Electronics + DAQ	\$3K
Calibration sources (Eu-152 ref)	\$1K
Total hardware	\$40-54K
Operator labor (6 mo)	(assume institutional)
Total cost	\$25-50K (excluding labor)

Comparison: \$50K is $\sim 1/100$ th of a LIGO-scale precision-physics experiment. Cheapest decisive BST test in the portfolio.

6. Timeline

Phase	Months	Activity
1	1-2	Cavity fabrication, source procurement, detector calibration
2	3-4	Initial paired counting, systematic characterization
3	5-6	Full alternation campaign (~672 hours counting)
4	7-9	Data analysis, systematic-drift verification
5	10-12	Replication + cross-check at second institution if positive
6	13-18	Manuscript preparation + peer review

Total: 6-18 months depending on replication requirement.

7. Falsification criteria

Positive result

- Measured τ_1/τ_0 in range [1.001, 1.003] at $\geq 3\sigma \rightarrow$ BST H4 CONFIRMED at predicted BST primary form
- Measured τ_1/τ_0 in range [1.001, 1.003] at higher precision $\rightarrow$ strengthens cross-anchor with Decca residual

Negative result

- Measured $|\tau_1/\tau_0 - 1| < 0.05\%$ at $\geq 5\sigma$ precision over 6 months counting $\rightarrow$ BST H4 FALSIFIED
- BST substrate-coupling ontology (per Paper #111) constrained: no decay-rate effect at boundary scale $\sim \mu\text{m}$
- Paths forward: extend to (a) longer $t_{1/2}$ isotopes (more sensitive), (b) cryogenic Casimir cavity (Sushkov-class), (c) different boundary geometries

Ambiguous result

- Measured ratio at $1-3\sigma$ deviation: requires replication at second institution + extended counting for resolution

8. Cross-anchor with SP-29 portfolio

If H4 confirms (positive result), the BST substrate-coupling factor 3/1507 is anchored at TWO independent measurements: - Casimir force residual (Decca 2007, already published, 0.2% measured) - Cs-137 decay rate (this experiment, predicted 0.2% measurable)

This makes BST substrate-coupling falsifiable in two domains at the same BST primary form. No standard model — Schwinger, Lifshitz, van-der-Waals — predicts the same factor in both contexts. A cross-anchor confirmation is BST-specific and observationally decisive.

If H4 refutes (null result), the BST hypothesis is constrained to “substrate-coupling modulates Casimir force residuals but NOT radioactive decay rates” — still consistent with the Decca observation but requires explaining why the same substrate-coupling factor appears in one phenomenon but not the other.

9. Companion: SP29-2 H1 falsifier

Per Lyra T2360 / Toy 3025 (2026-05-18): Sr clock measurement at $L = 100$ nm Casimir gap predicts $\Delta\nu/\nu \approx -4 \times 10^{-13}$, \$200-400K, 6-18 months. Independent decisive falsifier.

Combined SP29-1 + SP29-2 = two independent decisive BST tests for ~\$300K total. Either confirms substrate-coupling at gravitational scale; either falsifies cleanly. This represents the cheapest possible cross-domain BST validation in the entire literature.

10. Conclusion

SP29-1 (Cs-137 Casimir-boundary decay-rate modulation) is the highest-leverage single experimental test in BST's portfolio:

- Quantitative BST primary prediction: $\tau_1/\tau_0 = 1 + 3/1507 = 1.00199$
- Cross-anchored with Decca Casimir residual (same BST primary form)
- Decisive falsifiability at $\leq 0.05\%$ precision (vs predicted 0.2% signal)
- Cost \$25-50K (1/100 LIGO scale)
- Timeline 6-18 months
- No standard framework predicts any effect $\rightarrow$ BST signature unique

Status: paper-grade proposal v0.1 ready for experimental collaboration outreach. Suggested collaboration targets: precision γ -ray metrology groups at NIST, PTB, ENEA, or university nuclear physics departments with HPGe + Casimir capability.

Authors: Casey Koons (lead) with the Tekton CI collaboration: Elie (this proposal, Toys 3009/3020 Decca residual identification, BST primary prediction), Lyra (Paper #111 substrate dynamics, SP29-2 companion), Grace (null-model BST prediction calibration), Keeper (SP29-6 master ranking, K-audit), Cal (referee).

Proposal v0.1 filed 2026-05-18. Ready for Keeper K-audit + Cal referee review prior to experimental collaboration outreach.